

RaspbotV2 patrol car

1. Experimental preparation

1. Material preparation

- RaspbotV2 car *1
- Eight-way tracking module *1
- 7.4V battery *1
- Several Dupont wires
- Several M3 copper pillars and M3 screws

2. Car wiring

After assembling the car, it will look like the following figure



2.1 Wiring of Roboduino and infrared sensor (this example uses serial communication)

RaspbotV2	Infrared sensor
TX	RX
RX	TX
5V	5V
GND	GND

If the cable is not long enough, you can use a ch340 to connect to the USB port of the Raspberry Pi, and then connect the line tracking module to the ch340, and then comment line 17 of tracking.py and uncomment line 18.

```
1  #!/usr/bin/python
2  # -*- coding:utf-8 -*-
3  import serial
4  from Raspbot_Lib import Raspbot
5  import time
6  import PID
7
8  # Initialize pid
9  P = 6
10 I = 0
11 D = 0
12 middle_error = 0 #center
13 go_speed = 20
14 IR_track_PID = PID.PositionalPID(P, I, D) #PID parameters
15
16 #Open the serial port
17 ser = serial.Serial("/dev/ttyAMA0",115200,8,'N',1,timeout = 0.5) #Serial port of Raspberry Pi pin
18 #ser = serial.Serial("/dev/ttyUSB0",115200,8,'N',1,timeout = 0.5)
19
20 if __name__ == '__main__':
```

PID.py and tracking.py must be placed in the same folder in the car's image before they can be run. Otherwise, an error will be reported that the third-party library cannot be found in the following environment. You can install the corresponding library according to the information provided by the car to solve the error.

Main procedures

```
#Main function
if __name__ == "__main__":
    print("start it")

    try:
        while True:
            usart_deal()
    except KeyboardInterrupt:
        pass
    finally:
        #Stop Movement
        bot.Ctrl_Car(0,1,0) #L1 motor backward 0 speed
        bot.Ctrl_Car(1,1,0) #L2 motor backward 0 speed
        bot.Ctrl_Car(2,1,0) #R1 motor backward 0 speed
        bot.Ctrl_Car(3,1,0) #R2 motor backward 0 speed
        ser.write(bytes("$0,0,0#", 'utf-8'))
```

The main function is to perform PID processing for line patrol according to the value of the infrared probe, so that line patrol can be completed on the map with black lines and white background.

In tracking.py, there is a parameter for adjusting PID line patrol. If you want to increase or decrease the speed and optimize the effect, you can adjust the macro definition value inside.

```
#Initialize pid  
P = 6  
I = 0  
D = 0  
middle_error = 0 #center  
go_speed = 20
```

- P: P value of pid line patrol
- I: I value of pid line patrol
- D: D value of pid line patrol
- go_speed: speed of line patrol

Experimental phenomenon:

On the premise of ensuring that the wiring and installation are correct, after the 8-way line patrol module is calibrated, the following command can be used to start line patrol.

```
python3 tracking.py
```

If the 8-way module probe still cannot detect the black and white lines normally, you need to wait until the module works normally before starting the command.